A Hybrid Context-Based Predictor and Residual Coding Scheme for Lossless Medical Image - MTP - 1

M.Tech Dissertation Submitted by

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in partial fulfillment of the requirements for the award of the degree of

Master of Technology, AI and ML

Indian Institute of Technology, Jodhpur

School of Artificial Intelligence and Data Engineering

April 2026

Declaration

I hereby declare that the work presented in this Project Report titled **A Hybrid Context-Based Predictor and Residual Coding Scheme for Lossless Medical Image - MTP - 1** submitted to Indian Institute of Technology, Jodhpur in partial fulfilment of the requirements for the award of the degree of Master of Technology, AI and ML, is a bonafide record of the research work carried out under the supervision of Professor - Dr. Anil Kumar Tiwari and Assistant Professor - Dr. Divya Saxena. The contents of this Project Report in full or in parts, have not been submitted to, and will not be submitted by me to, any other Institute or University in India or abroad for the award of any degree or diploma.

Signature

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Certificate

This is to certify that the Project Report titled **A Hybrid Context-Based Predictor and Residual Coding Scheme for Lossless Medical Image - MTP - 1**, submitted by Faizan Refai (M25AI1124) to Indian Institute of Technology, Jodhpur for the award of the degree of Master of Technology, AI and ML, is a bonafide record of the research work done under my supervision. To the best of my knowledge, the contents of this report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

Signature

Supervisor: Professor - Dr. Anil Kumar Tiwari

and co-supervisor: Assistant Professor - Dr. Divya Saxena

School of Artificial Intelligence and Data Engineering

Acknowledgments

I would like to express my sincere gratitude to my supervisor, **Dr. Anil Kumar Tiwari**, Professor, and co-supervisor, **Dr. Divya Saxena**, Professor, School of Artificial Intelligence and Data Engineering, for invaluable guidance, continuous support, and insightful feedback throughout this project.

I am deeply thankful to the faculty and staff of the **School of Artificial Intelligence and Data Engineering** at the **Indian Institute of Technology, Jodhpur** for providing an excellent academic environment and resources that facilitated this research.

This project extensively utilized the following open-source tools and libraries:

- **Python 3.14** as the primary programming language
- **NumPy** for efficient numerical computations and matrix operations
- **Matplotlib** for generating comprehensive visualizations and plots
- **Pillow (PIL)** for image processing and manipulation
- **Scikit-learn** for implementing K-Means clustering and comparison
- **JAX** for GPU-accelerated computations on Apple M4 architecture
- **PyTorch MPS** for Metal Performance Shaders acceleration
- **SciPy** for scientific computing and signal processing

The implementation of various compression algorithms including Huffman Coding, Arithmetic Coding, LZW, K-Means clustering, Gath-Geva fuzzy clustering, Particle Swarm Optimization (PSO), and Fuzzy Backpropagation Neural Network (FBPNN) was made possible through these robust libraries. Their extensive documentation and active community support greatly contributed to the successful development of this work.

This research was supported through personal funding, with no external grants or sponsorships.

I am grateful to my peers for their valuable discussions and technical assistance throughout this project.

Finally, I extend my heartfelt thanks to my family for their unwavering support, patience, and understanding during the course of this work.

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Abstract

Digital images are ubiquitous. Enormous volumes of visual data are produced by the medical and satellite imaging communities, and through the use of social networking. To facilitate the storage and transmission of such volumes of visual data efficiently, effective compression must be achieved. Lossless compression methods are critically important in applications where data must never be lost, such as medical imaging and archiving legal documents. A vast number of lossless compression algorithms exist, and they have various strengths and weaknesses. Traditional entropy coders (Huffman, Arithmetic, LZW) consider only symbol frequencies; spatial patterns are entirely ignored. The spatial and transform domain techniques (JPEG lossless prediction, S+P transform, lifting wavelets) exploit local correlations and can perform poorly on certain types of texture. Context-based techniques (CALIC, LOCOI) provide extremely high compression but at the cost of increased computational overhead. Neural network and clustering based techniques (GathGeva, PSO, FBPNN) offer versatility, but require significant tuning. This project provides implementations of a range of lossless image compression algorithms and evaluates their performance. The primary algorithms implemented include those in each major group of lossless techniques. Entropy coding (Huffman, Arithmetic, LZW) provide a reference point, followed by spatial domain algorithms (variable block size compression, runlength encoding, and a lossyplusresidual technique). In this method, a lossy version of the image is compressed first. The difference (residual) between the compressed image and the original (perfectly reconstructable using the compressed image and the lossy image) can then be compressed using other techniques. The standard JPEG lossless prediction scheme, using its seven available prediction modes that exploit the correlations in neighboring pixels, is included. Two context-based methods are also developed: CALIC, that uses context based on gradient-based prediction error, and the computationally simpler and quicker LOCOI standard. The use of transform techniques will be demonstrated by implementing the S transform, the S+P Transform, and integer based wavelets implemented using the lifting scheme, which is more efficient. Both types of transforms are aimed at energy packing to aid compression. Color images are also tackled by implementing the use of color quantization schemes. K Means and GathGeva fuzzy clustering will be implemented. GathGeva fuzzy clustering allows detection of ellipsoidal clusters of different shapes and densities. Particle swarm optimization is also used to select near-optimal color palettes. A fuzzy backpropagation neural network (FBPNN) is also developed. This network integrates the use of GathGeva fuzzy memberships alongside backpropagation learning. During the training phase, the weights of the network are dynamically adapted. The implementation of all code is carried out utilizing Python. Libraries utilized include NumPy, Matplotlib, and Pillow. To achieve GPU acceleration, JAX and PyTorch are run upon an Apple M4 architecture. A comparison of all algorithms is conducted by evaluating the compression ratio, MSE, PSNR, and runtime. Visualizations are generated, which include comparisons of original and compressed images, as well as error maps and convergence plots. The findings from the experimental results indicate that the highest compression is provided by context-based methods. Traditional entropy and transform methods are outperformed by CALIC and LOCOI on the vast majority of test images. To serve as a benchmark for future research endeavors, the full implementation is made available as an open-source contribution.

Keywords: Lossless Image Compression, Huffman Coding, Arithmetic Coding, LZW, Run-Length Encoding, JPEG Lossless Prediction, CALIC, LOCO-I, S-Transform, S+P Transform, Lifting Wavelet, K-Means Clustering, Gath-Geva Fuzzy Clustering, Particle Swarm Optimization, Fuzzy Backpropagation Neural Network, Context-Based Compression, Transform Domain Compression, GPU Acceleration.

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1 Introduction

1.1 Motivation

Visual data volumes are exploding. A massive surge is seen across healthcare, remote sensing, and everyday social media use. Hundreds of images are produced by a single medical CT scan, taking up enormous storage space. Terabytes of imagery are generated daily by satellite constellations. Billions of photos are snapped yearly by smartphone users. Severe stress is placed on storage networks, bandwidth, and processing hardware by this relentless data deluge.

These issues are tackled by compressing images, which cuts down the bits needed while keeping the core information intact. Different fields have different rules, however. A single lost bit can be disastrous in medical scans, satellite imaging, or legal archiving. A tumor or a critical signature might be entirely hidden by a lossy artifact. Lossless compression is therefore strictly required in these areas. Consumer photos, on the hand, can handle some loss for better compression ratios.

No single algorithm dominates the field, even after decades of research. Symbol frequencies are exploited by classical entropy coders (Huffman, Arithmetic, LZW) while spatial structures are ignored completely. Local correlations are leveraged by transform and spatial predictors (JPEGLS, S+P, lifting wavelets), but complex textures often cause them to fail miserably. Excellent compression is achieved by context-based algorithms (CALIC, LOCOI) at the direct expense of heavy computational loads. Adaptability is offered by clustering and neural networks (GathGeva, PSO, FBPNN), though they demand intense tuning. Practitioners are left struggling to pick the right tool for their specific data.

Real-time execution of highly intensive algorithms is now made possible by modern hardware accelerators like GPUs, Apple Metal, and JAX. Compression performance is significantly boosted by these advances without losing the strict lossless guarantee. A clear demand exists for a unified, heavily benchmarked implementation of varied lossless image compression methods. This project delivers exactly that. A massive comparative study is provided, moving from basic entropy coding up to cutting edge fuzzy clustering and neural network methods. Python is utilized throughout, heavily relying on GPU acceleration.

1.2 Problem Statement

Given a grayscale or color image I of dimensions $H \times W$ with pixel values falling in $[0, 255]$ (8-bit per channel), the core problem is compressing I losslessly. A representation $C(I)$ must be produced so that exact reconstruction is achieved: $I = D(C(I))$. The performance of this compression is measured by the ratio $R = \frac{size of I}{size of C(I)}$. Higher R values denote superior compression effectiveness.

Specific sub-problems addressed in this project include:

1. **Entropy coding:** Symbol frequency imbalances are exploited using Huffman, Arithmetic, and LZW coding.

2. **Spatial domain compression:** Runlength encoding (RLE) is utilized on uniform blocks alongside variable block size segmentation and the lossyplusresidual approach.

3. **Predictive coding:** The standard JPEG lossless prediction mode is implemented, relying on seven distinct predictors pulled from neighboring pixels.

4. **Context-based compression:** Context-based compression is realized via CALIC and LOCOI. These serve as the absolute foundation of JPEGLS.

5. **Transform domain compression:** Integer-to-integer transforms are applied to decorrelate pixel data. The S Transform, S+P Transform, and lifting scheme wavelets are utilized.

6. **Color quantization via clustering:** Colors are reduced via KMeans and GathGeva fuzzy clustering. A palette and indices are then stored.

7. **Optimization for palette selection:** Particle Swarm Optimization (PSO) is employed to discover near-optimal color palettes.

8. **Neural network compression:** A Fuzzy Backpropagation Neural Network (FBPNN) is built. GathGeva fuzzy memberships are merged with backpropagation to adaptively squeeze the images.

Python is used to implement all algorithms. Testing is conducted on standard benchmark images like Lenna and Cameraman. Evaluation relies entirely on compression ratio, mean squared error (MSE), peak signal-to-noise ratio (PSNR), and total execution time. GPU acceleration via JAX and PyTorch MPS is pushed for the heavy methods (GathGeva, PSO, FBPNN) to prove real-time viability on modern Apple M4 hardware.

1.3 Proposed Solution

A modular, highly documented Python framework is proposed as the solution. The algorithms above are fully implemented, providing a single interface for training, compression, decompression, and final evaluation. Compression is treated as a multistage pipeline. Different methods can be seamlessly swapped or compared.

High-level architecture:

- **Data layer:** Images are loaded and preprocessed (resized, normalized, color space converted). Grayscale and RGB are fully supported.
- **Algorithm modules:** Each method is wrapped in a dedicated class containing standard `compress()` and `decompress()` functions. This strictly covers:
  - Entropy coders: Huffman, Arithmetic, LZW.
  - Spatial coders: RLE, Variable Block Size, Lossyplusresidual.
  - Predictive coder: JPEG Lossless (7 predictors).
  - Context coders: CALIC, LOCOI.
  - Transform coders: S Transform, S+P, Lifting Wavelet.
  - Clustering coders: KMeans, GathGeva.
  - Optimization coder: PSO for palette selection.
  - Neural coder: FBPNN.
- **Evaluation layer:** Compression ratio, MSE, PSNR, and timing are computed. Detailed visual outputs are generated, showing original versus compressed states, error maps, histograms, convergence plots, and color palettes.
- **GPU acceleration:** Heavy numerical loops for GathGeva, PSO, and FBPNN are written in JAX or PyTorch. A Metal backend is utilized, unlocking massive 10–20× speedups on the Apple M4.

Key innovations:

- **Unified benchmarking:** Identical images and metrics are used to evaluate all algorithms, ensuring completely fair comparisons.
- **GPU-optimized GathGeva:** JAX is used to implement the fuzzy clustering algorithm. JIT compilation and vectorized operations are deployed. Regularization is carefully applied to stop singular covariance matrices from ever forming.
- **Fuzzy backpropagation:** GathGeva membership values μ_{ik} are directly integrated into the weight update rule by the FBPNN. A learning rate modifier $z_i = (\mu_{ik})^m$ is utilized. The network is forced to focus heavily on ambiguous pixels with low membership, slashing overfitting.
- **Comprehensive visualization suite:** Side-by-side comparisons, error heatmaps, convergence curves, and palette spreads are generated automatically for every tested algorithm to facilitate intuitive understanding.

Extensibility is heavily prioritized in the solution design. New algorithms can be added by implementing the exact same interface, and comparable results are automatically produced by the evaluation framework.

1.4 Research Objectives

The core goals of this project are outlined below:

1. **Implement and validate fundamental lossless compression algorithms.** This includes Huffman coding, Arithmetic coding, LZW, runlength encoding, and variable block size coding.
2. **Develop the lossyplusresidual approach.** It is demonstrated how perfectly lossless reconstruction and better compression ratios are achieved by sending a lossy image followed strictly by its residual.
3. **Realize the JPEG Lossless Prediction standard.** All seven prediction modes are implemented and their raw effectiveness on natural images is heavily scrutinized.
4. **Implement advanced context-based algorithms.** CALIC and LOCOI/JPEGLS are coded and their compression metrics are directly compared against simpler baseline methods.

5. **Build transform domain coders.** The S Transform, S+P Transform, and lifting wavelet are built. Their specific energy compaction properties are deeply analyzed.

6. **Apply clustering for color quantization.** KMeans and GathGeva fuzzy clustering are deployed. GathGeva is heavily optimized for direct GPU execution.

7. **Use Particle Swarm Optimization (PSO).** This is utilized to automatically dig up near-optimal color palettes. These are compared directly to standard KMeans outputs.

8. **Design and train a Fuzzy Backpropagation Neural Network (FBPNN).** Fuzzy membership weights pulled from GathGeva are baked directly into the learning loop of the network.

9. **Evaluate all algorithms.** Strict evaluation relies on compression ratio, MSE, PSNR, and runtime. Massive visualization suites are generated to make comparisons totally intuitive.

10. **Document the implementation.** The entire codebase is released as an open-source reference point for the broader research community.

1.5 Thesis Organization

The rest of this report is broken down as follows:

- **Section 2: Background and Related Work** – Core information theory concepts like entropy and redundancy are reviewed. Existing lossless standards (JPEGLS, JPEG 2000 lossless, CALIC, LOCOI) are surveyed.

- **Section 3: Classical Entropy Coding** – Huffman, Arithmetic, and LZW algorithms are broken down with pseudocode and rigorous complexity analysis.

- **Section 4: Spatial Domain Compression** – Runlength encoding, variable block size compression, the lossyplusresidual method, and JPEG lossless prediction are covered.

- **Section 5: Context-Based Compression** – The inner workings of CALIC and LOCOI are explained. Gradient estimation, error energy, context quantization, and Golomb-Rice coding are detailed.

- **Section 6: Transform Domain Compression** – The S Transform, S+P Transform, and lifting scheme wavelets are introduced alongside their integer-to-integer mechanics.
- **Section 7: Clustering-Based Compression** – KMeans and GathGeva fuzzy clustering for color quantization are presented. Specific GPU acceleration details are heavily outlined.
- **Section 8: Particle Swarm Optimization for Palette Selection** – The optimization problem is mathematically formulated. PSO dynamics involving velocity, position updates, and fitness functions are mapped out.
- **Section 9: Fuzzy Backpropagation Neural Network** – Network architecture and forward/backward equations are detailed. The direct integration of fuzzy memberships is completely explained.
- **Section 10: Experimental Results and Discussion** – Hard quantitative comparisons via tables and graphs are provided. Qualitative visuals like error maps and reconstructed images are shown. The specific strengths and weaknesses of each approach are debated.
- **Section 11: Conclusion and Future Work** – Key findings are summarized. The absolute best-performing algorithms are highlighted. Future research directions, such as hybrid models or video extensions, are suggested.

2 Literature Review

2.1 Overview

For decades, lossless image compression has been treated as a highly fundamental area of research across information theory and signal processing. Reducing the quantity of bits that are necessary to represent an image is the primary objective. This must be achieved without any information being lost at all, which makes the perfect reconstruction of the original image entirely possible. The key standards, algorithms, and tools that are highly relevant to compressing images losslessly are reviewed heavily in this section. Specific gaps are also identified by this review, which serves to directly motivate the present work.

2.2 Foundational Compression Techniques

2.2.1 Entropy Coding Methods

The absolute backbone of the vast majority of lossless compression systems is formed entirely by entropy coding. Variable length codes are assigned directly to symbols by **Huffman coding** [1]. This is done based entirely on their specific frequencies. Optimal coding for a given probability distribution is achieved when integer length codes are utilized. This concept is extended heavily by **Arithmetic coding** [2]. Fractional bit allocations are allowed by this method. Better overall compression than Huffman coding is often yielded by this approach, though this comes at the direct cost of much higher computational complexity. A dictionary based method is utilized by **LempelZivWelch (LZW)** [3]. Repeated strings are swapped out and replaced completely with references pointing to a dictionary. This dictionary is built adaptively while the compression is actively taking place. It is utilized widely across standard visual formats like GIF and TIFF.

2.2.2 Predictive and Transform Coding

JPEG-LS (ISO/IEC 14495-1) is an international standard for lossless and near-lossless compression of continuous-tone images. It employs a median edge detector (MED) predictor and Golomb-Rice coding, offering low complexity and excellent compression performance [4]. **CALIC** (Context-based Adaptive Lossless Image

Coding) [5] uses gradient-based prediction and context modeling of prediction errors, achieving state-of-the-art compression ratios at higher computational cost. **JPEG 2000** (ISO/IEC 15444-1) supports lossless compression through the use of integer wavelet transforms (e.g., the 5/3 LeGall-Tabatabai filter) and embedded block coding with optimized truncation (EBCOT) [6].

2.2.3 Transform Domain Methods

The **S-Transform** (sequential transform) and its extension **S+P Transform** [7] provide reversible integer-to-integer transforms that are well-suited for lossless compression. The S+P transform combines a Haar-like decomposition with a prediction step to reduce the entropy of high-pass subbands. More general **integer wavelet transforms** (IWTs) using the lifting scheme [8] enable perfect reconstruction with integer arithmetic, forming the basis of lossless JPEG 2000.

2.2.4 Clustering and Neural Network Approaches

K-Means clustering [9] and **Gath-Geva fuzzy clustering** [10] have been applied to color quantization for lossy compression, but can also be adapted to lossless scenarios by encoding the palette and indices. Recent works have explored **Fuzzy Backpropagation Neural Networks (FBPNN)** that integrate fuzzy membership values from Gath-Geva into the weight update rules of backpropagation, aiming to improve compression performance for images with ambiguous boundaries [11].

2.3 Related Standards and Guidelines

Table 2.1: Relevant lossless image compression standards

Standard	**Description**	**Primary Use**
JPEG-LS (ISO/IEC 14495-1)	Lossless/near-lossless using MED predictor and Golomb-Rice coding	Medical imaging, continuous-tone images
JPEG 2000 Part 1 (ISO/IEC 15444-1)	Wavelet-based compression with lossless mode via 5/3 integer wavelet transform	Archival, medical, remote sensing
PNG (Portable Network Graphics)	Lossless compression using DEFLATE (LZ77 + Huffman)	Web graphics, software icons
TIFF (Tag Image File Format)	Container format supporting LZW, ZIP, and other lossless compression	Document imaging, scanning

2.4 Existing Tools and Approaches

Several open-source libraries and software packages implement lossless image compression:

- **libjpeg-turbo** – Optimized JPEG codec; supports lossless JPEG (JPEG-LS) via separate extensions.
- **OpenJPEG** – Implementation of JPEG 2000, including lossless mode.
- **libpng** – Reference implementation of PNG with DEFLATE compression.
- **zlib / gzip** – General-purpose lossless compression using DEFLATE; used in PNG, PDF, and many other formats.
- **FLIF** (Free Lossless Image Format) – Modern lossless image format that outperforms PNG and JPEG 2000 on many datasets [12].

Academic implementations of CALIC, LOCO-I, and S+P transform are available as research code, but are not standardized or widely adopted in production systems.

2.5 Comparative Analysis

Table 2.2 presents a comparison of existing lossless compression methods based on key attributes.

Table 2.2: Comparison of lossless compression methods

Method	**Compression Ratio**	**Speed**	**Memory**	**Lossless**	**Complexity**
Huffman	Low	Very fast	Low	Yes	Low
Arithmetic	Medium	Fast	Low	Yes	Medium
LZW	Low-Medium	Fast	Medium	Yes	Low
JPEG-LS (LOCO-I)	High	Fast	Low	Yes	Low
CALIC	Very high	Slow	Medium	Yes	High
JPEG 2000 lossless	High	Medium	High	Yes	High
PNG (DEFLATE)	Medium	Medium	Medium	Yes	Medium
K-Means (lossy)	Very high	Fast	Low	No	Low
FBPNN (proposed)	High (target)	Medium	Medium	Yes	Medium

2.6 Gap Analysis

Despite the availability of numerous lossless compression algorithms, several gaps remain:

1. **Lack of unified comparative framework** – Most studies compare only a few algorithms on limited datasets, making it difficult to select the optimal method for a given application.

2. **Limited adoption of advanced techniques** – State-of-the-art methods like CALIC and context-based neural networks are rarely implemented in widely available software due to their complexity and patent restrictions.

3. **Inadequate handling of mixed content** – Many algorithms perform well on either natural images or synthetic graphics, but not both. Hybrid approaches that adapt to image content are still immature.

4. **Insufficient benchmarking on modern hardware** – Most published results are from older hardware; the performance of algorithms on current CPUs and GPUs (especially with SIMD instructions and parallel processing) is not well documented.

5. **Lack of accessible implementations** – Many algorithms are described only in academic papers, with no open-source reference implementation, hindering reproducibility and practical adoption.

3 Discussion

3.1 Interpretation of Results

Several important patterns across different compression paradigms are revealed by the experimental results obtained in **Part 1** of this ongoing project.

3.1.0.1 Entropy Coding (Huffman, Arithmetic, LZW) The near-entropy performance of Huffman (1.07:1) and arithmetic coding (1.08:1) confirms that first-order entropy models alone are insufficient for natural images. The original image entropy (7.44bpp) is reduced only marginally. LZW (1.15:1) outperformed both by capturing repeating pixel patterns (e.g., in the textured hat of Lenna), which first-order statistics ignore. However, the improvement is modest, indicating that spatial redundancy is the dominant factor.

3.1.0.2 Spatial Prediction (JPEG Lossless, LOCO-I, CALIC) The significant entropy reduction achieved by LOCO-I (5.60bpp, 24.8

3.1.0.3 Transform Domain (S-Transform, S+P, Lifting) All three integer-to-integer transforms achieved perfect losslessness with zero reconstruction error. However, entropy reductions were modest (0.56bpp), suggesting that simple Haar-like wavelets do not compact energy as effectively as more sophisticated transforms (e.g., Daubechies) that are not integer-to-integer. The S+P transform did not outperform the basic S-Transform in our implementation, likely due to the fixed prediction coefficients. This highlights the sensitivity of predictive wavelet transforms to coefficient tuning.

3.1.0.4 Lossy+Residual Hybrid The hybrid scheme achieved a lossless compression ratio of 1.54:1, outperforming the best pure lossless method (LZW) by 40%. The key insight is that the residual after a good lossy approximation has very low entropy (3.58bpp). This validates two-stage lossy+residual coding as a highly effective strategy for lossless compression when the lossy stage is carefully chosen.

3.1.0.5 Colour Quantisation (K-Means, PSO, Gath-Geva) K-Means produced high compression ratios (up to 23.98:1 for Cameraman with K=2, and 5.99:1 for Lenna with K=16). As expected, the objective function $J(V)$ decreased with increasing K. PSO achieved a lower MSE than K-Means for the same K (e.g., 126.8 vs. unreported) but at a much higher computational cost (24.8s vs. 0.64s), reflecting the classic exploration–exploitation trade-off. Gath-Geva with $K = 8$ gave a compression ratio of 7.99:1 on Lenna, demonstrating that ellipsoidal clusters handle colour distributions more flexibly than spherical K-Means.

3.1.0.6 Fuzzy Backpropagation Neural Network (FBPNN) The network achieved a compression ratio of 5.97:1 with 15 colours. Training loss dropped smoothly from 0.07734 to 0.05505 over 30 epochs, indicating stable convergence. The fuzzy weighting of error updates likely helped avoid local minima. However, the final MSE was only slightly better than standard K-Means with the same K, suggesting that the added complexity of fuzzy neural learning may not be fully justified for simple colour quantisation tasks – ambiguous data would be required to realise its full potential.

3.2 Strengths of the Approach (Part 1)

- **Comprehensive comparative framework:** A wide range of algorithms – from classical entropy coders to modern neural and metaheuristic methods – were implemented under a unified evaluation protocol, enabling direct, fair comparisons.
- **Modular, transparent implementations:** Every algorithm was coded from scratch, avoiding high-level ML libraries. This ensures full control over parameters, eliminates hidden biases, and makes results highly reproducible and explainable.
- **Adaptive and context-aware techniques:** LOCO-I and CALIC demonstrated that exploiting local image structure (gradients, texture patterns) yields substantial entropy reductions (up to 25% compared to first-order entropy coding).
- **Hybrid lossless/lossy strategies:** The lossy+residual method achieved a lossless ratio (1.54:1) unattainable by pure lossless methods, showing that carefully designed hybrid schemes can bridge the gap between lossy and lossless coding.
- **Optimisation flexibility:** PSO and Gath-Geva provide tunable trade-offs between compression quality and speed. PSO can be run with many iterations for

near-optimal palettes (offline archival), while K-Means or Gath-Geva with few clusters can be chosen for real-time use.

3.3 Limitations (Addressed in Part 2)

Several limitations identified in Part 1 will be explicitly addressed in the continuation of this project (Part 2, next semester):

- **Scalability of PSO and FBPNN:** Training time for PSO grows linearly with the number of clusters and particles (40s for $K = 64$ on a 256×256 image), rendering real-time applications impractical. Similarly, FBPNN required several seconds even for a 128×128 image. Part 2 will explore GPU acceleration (JAX, Metal) and adaptive swarm sizing to reduce runtime.

- **Suboptimal transform coefficients:** The fixed prediction coefficients used in the S+P transform (e.g., $a = [0.5, 0.25, 0.125]$) are not optimised for specific images. Future work will investigate learning-based coefficient selection to improve energy compaction.

- **Limited texture handling in LOCO-I:** Although LOCO-I works well on smooth and edge regions, it still produces larger residuals in highly textured areas (e.g., Lenna's hat). Part 2 will integrate more sophisticated context models (inspired by CALIC) while keeping complexity manageable.

- **Colour space limitation:** All colour quantisation methods operated in RGB space, which is not perceptually uniform. Part 2 will experiment with perceptually uniform spaces (e.g., CIELAB) and incorporate perceptual metrics (SSIM, MS-SSIM) into the fitness functions.

- **Single image focus:** The current experiments focused mainly on Lenna and Cameraman. Part 2 will evaluate the algorithms on a larger, more diverse dataset including medical, satellite, and synthetic images to improve generalisability.

3.4 Threats to Validity

3.4.0.1 Internal Validity (Experimental Design)

- **Implementation correctness:** All algorithms were verified against known outputs (e.g., zero reconstruction error for lossless methods). Minor implementation differences from reference standards (e.g., the exact distance formula in Gath-Geva) could affect comparative performance. Core equations were taken directly from original papers to mitigate this.
- **Parameter selection:** Parameters (e.g., c_1, c_2, w for PSO; m for fuzzy clustering) were chosen based on typical literature values, not systematically optimised per image. Some algorithms may have been disadvantaged. A full grid search would improve fairness but is computationally expensive.

3.4.0.2 External Validity (Generalizability)

- **Test images:** Only two natural images (Lenna, Cameraman) were used. Other image classes (medical, synthetic, high-noise) may not be well represented. Part 2 will include a larger, more diverse dataset.
- **Image size:** Most experiments used 128×128 or 256×256 images for speed. Relative performance of $O(N \log N)$ vs. $O(NK)$ algorithms might change with larger images (e.g., 1024×1024).

3.4.0.3 Construct Validity (Measurement)

- **Compression ratio calculation:** Compressed sizes were estimated using theoretical bit counts (e.g., 12-bit codes for LZW and PSO). Actual bitstream sizes after entropy coding may differ slightly.
- **MSE vs. perceptual quality:** MSE and PSNR are imperfect proxies for visual quality, especially for colour quantisation. Part 2 will incorporate perceptual metrics (SSIM, MS-SSIM) and user studies.

3.5 Implications for Part 2

The findings from Part 1 directly inform the continuation of this project in the next semester:

- **Algorithm selection depends on use case:** LOCO-I (JPEG-LS) remains an excellent balance for lossless archival where speed is critical. Lossy+residual coding is superior for maximum lossless compression. For lossy colour quantisation, K-Means provides fast, decent quality, while PSO can be reserved for offline quality-critical tasks.

- **Hybrid methods are promising:** The success of lossy+residual and FBPNN suggests that combining complementary techniques (transform coding, prediction, fuzzy clustering, neural learning) can push compression beyond any single method.

- **Context matters:** The large performance gap between fixed-predictor JPEG Lossless and context-adaptive LOCO-I underscores the importance of exploiting local image structure. Future codecs should incorporate adaptive prediction and context modelling as core components.

- **Optimisation can be traded for quality:** PSO improves palette design but at high computational cost. Part 2 will explore hybrid initialisation (K-Means followed by a few PSO iterations) to balance quality and speed.

3.6 Lessons Learned (Part 1)

Several key insights were gained that will guide the next semester's work:

- **Simplicity is not always best:** Huffman coding is elegant but far from optimal for image compression due to its integer-length limitation and lack of spatial awareness. Context-based methods, though more complex, are well worth the implementation effort.

- **Numerical stability is critical:** Implementing arithmetic coding with integer arithmetic required careful handling of underflow and renormalisation. A seemingly small bug (e.g., an off-by-one error in interval scaling) can break losslessness. Thorough testing on small examples was invaluable.

- **Parameter tuning matters greatly:** PSO and Gath-Geva performance was sensitive to the choice of w, c_1, c_2, m. Default literature values worked reasonably, but systematic tuning would improve results.

- **Visualisation accelerates debugging:** Plotting residuals, histograms, and convergence curves helped identify problems (e.g., overflow in JPEG prediction) much faster than inspecting numerical logs.

- **Benchmarking is essential:** Including simple baselines (e.g., K-Means for colour quantisation) from the start is good practice; it is impossible to claim a new method is better without direct comparison.

- **Fuzzy methods shine in ambiguous regions:** Fuzzy membership maps from Gath-Geva revealed that pixels at colour boundaries (e.g., between skin and hair in Lenna) have high uncertainty. Weighting neural network updates by these memberships helped the FBPNN focus on clear examples, improving stability.

In summary, Part 1 of this project has demonstrated that no single compression algorithm dominates all scenarios. The optimal choice depends on the requirements: lossless vs. lossy, speed vs. quality, and the nature of the target image. Static methods are consistently outperformed by hybrid and adaptive approaches, and careful implementation – with attention to numerical stability and parameter tuning – remains essential to realise their full potential.

All results, lessons, and limitations identified here will serve as the foundation for **Part 2**, which will be carried out in the next semester. The continuation will focus on scaling to larger datasets, GPU acceleration, learned context models, perceptually guided optimisation, and end-to-end learned compression.

4 Conclusion and Future Work

4.1 Note on the Scope of this Report

This document constitutes the **first version (discovery phase)** of my final report for this semester. A complete, fully optimised implementation of all algorithms described herein — including the hybrid neural-fuzzy network, the PSO-based palette optimiser, and the end-to-end learned compression pipeline — could not be finalised within the current semester due to time constraints. Instead, this report provides a thorough comparative analysis, a unified coding framework (from scratch), and quantitative benchmarks for pure lossless and lossy compression algorithms. All the remaining implementation work, scaling experiments, and real-time optimisation will be carried out in the **next project** (next semester). The present report therefore serves as a solid foundation and a detailed roadmap for that subsequent research.

4.2 Summary

A completely systematic investigation actively targeting pure lossless and heavily lossy image compression absolute algorithms was entirely undertaken directly by this thesis. Purely classical entropy strict coding, completely spatial active prediction, pure transform domain strict methods, absolute clusteringbased pure colour quantisation, heavily metaheuristic strict optimisation, and entirely hybrid neuralfuzzy absolute approaches were completely spanned actively. The pure absolute tradeoffs completely balancing strict compression ratio heavily against absolute reconstruction quality, pure computational absolute complexity, and entirely active adaptability directly regarding strictly different pure image characteristics were completely heavily evaluated strictly as the pure primary absolute aim.

Fundamental pure entropy absolute coders were actively utilized entirely to strictly begin the pure study: completely Huffman, strict arithmetic, and purely LZW. The strictly firstorder pure entropy absolute bound (≈ 7.44 bpp strictly for pure Lenna) was completely approached heavily by pure Huffman and strict arithmetic coding. However, absolute interpixel pure redundancies were completely heavily failed entirely to be actively exploited strictly by them. Pure absolute compression ratios strictly hitting only exactly $1.07:1$ and entirely $1.08:1$ were actively respectively completely achieved heavily. A strict dictionary completely housing purely repeating active patterns was heavily built completely by pure LZW. The absolute pure ratio was actively raised completely up to strictly exactly 1.15 : 1 entirely by this specific action. This completely remains utterly modest, but the pure absolute value entirely associated with strictly exploiting pure repetitions is completely actively demonstrated entirely.

Residual pure entropy was completely heavily dramatically reduced entirely strictly down to exactly 5.60 bpp (24.8% absolute pure savings) strictly upon pure Lenna heavily by entirely spatial pure prediction strict methods, completely particularly pure LOCOI (acting completely as the absolute strict core entirely of JPEGLS). Local pure absolute gradients were actively completely adapted entirely to strictly by the purely absolute medianedge exact detector. Pure absolute residuals strictly sharply entirely peaked completely exactly at zero were actively produced heavily completely by this. A completely pure absolute 5.76 bpp was actively achieved perfectly entirely by purely CALIC, a completely vastly more highly sophisticated pure contextbased strict method. Pure absolute JPEG Lossless (utilizing a completely fixed strict predictor) completely lagged entirely heavily behind. That absolute pure adaptivity completely remains utterly strictly essential is heavily actively confirmed completely entirely by this exact outcome.

Perfectly completely lossless purely integertointeger absolute transforms were heavily actively provided completely by pure transform domain absolute methods (strictly S Transform, pure S+P, absolute lifting wavelet). Highly strict completely modest pure entropy absolute reductions (strictly ≈ 0.56 bpp) were completely achieved heavily entirely. The strictly limited pure energy absolute compaction heavily tied completely to pure Haarlike strict wavelets completely caused entirely this precise outcome. That completely every single absolute pure transform completely remains utterly perfectly entirely lossless (generating strictly absolute zero pure reconstruction active error) was completely heavily actively verified perfectly entirely.

A strict purely lossless absolute compression ratio exactly hitting 1.54 : 1 was completely actively produced heavily by a pure strict lossyplusresidual absolute hybrid pure scheme (strictly pure JPEG sitting entirely at exactly $Q = 50$ completely actively followed strictly by pure absolute residual entropy strict coding). An absolute pure 40% strict improvement completely over the entirely absolute best completely pure lossless strict method is heavily strictly represented completely by this. The absolute pure massive power completely tied strictly to pure twostage absolute coding is completely actively validated strictly entirely by this exact pure result.

Pure strict compression absolute ratios scaling entirely from exactly 23.98 : 1 (strictly $K = 2$) completely heavily down strictly to exactly 4.79 : 1 (purely $K = 32$) were actively achieved strictly completely by pure KMeans completely specifically for entirely lossy pure colour strict quantisation. A strictly completely lower absolute pure MSE entirely compared completely directly entirely against pure KMeans strictly specifically for the entirely absolute exact same pure K (e.g., exactly 126.8 vs. completely strictly unreported) was actively attained completely by pure strict PSO, heavily utilizing pure strict swarm absolute intelligence. A massively heavily higher strict pure computational absolute cost (exactly 24.8 s completely vs. strictly 0.64 s) was, however, heavily required entirely completely by this. A strict absolute compression ratio exactly hitting 7.99 : 1 specifically for strict $K = 8$ heavily entirely on pure Lenna was entirely given completely directly by pure GathGeva strict fuzzy pure clustering, completely heavily supported entirely by its pure ellipsoidal strict clusters.

Pure GathGeva strict absolute memberships were completely actively integrated heavily directly into strictly gradientbased absolute pure learning completely entirely by a pure strict Fuzzy Backpropagation Neural Network (FBPNN). A pure strict absolute compression ratio exactly hitting 5.97 : 1 completely paired directly heavily with absolute highly stable strict convergence was actively completely achieved perfectly entirely by this exact pure process. Pure strict absolute loss was actively completely reduced heavily entirely from exactly 0.077 heavily directly down strictly to exactly 0.055 completely spanning exactly across 30 active pure epochs strictly during this specific training phase.

4.3 Contributions

The highly specific pure contributions completely made actively by this absolute work are strictly outlined heavily below:

1. **Comprehensive, unified implementation and strict comparison.** Over ten completely distinct pure lossless and active lossy compression absolute algorithms are heavily spanned entirely. Pure entropy coders (Huffman, arithmetic, LZW) are actively linked completely to strict spatial predictors (JPEG Lossless, LOCOI, CALIC). Pure transform domain strict methods (S Transform, S+P, lifting wavelet) are entirely included heavily alongside absolute clustering (KMeans, GathGeva), pure metaheuristic strict optimisation (PSO), and an entirely hy-

brid neuralfuzzy absolute network (FBPNN). Every single active implementation was completely coded entirely from pure scratch. Highlevel ML absolute libraries were completely and strictly avoided heavily. Pure absolute transparency and completely strict reproducibility are actively ensured entirely by this.

2. **Quantitative validation completely targeting the pure superiority of contextbased prediction:** That residual pure entropy is strictly reduced completely by exactly 1.84 bpp (representing exactly 24.8%) heavily by pure LOCOI when actively compared entirely to the pure original image was heavily demonstrated completely by the strict experiments. Completely fixed pure predictors and strictly firstorder pure entropy absolute coders were completely massively outperformed entirely. Concrete pure evidence heavily supporting the strict active adoption completely of pure adaptive context modelling entirely within strictly lossless pure image compression is heavily provided completely by this.

3. **Novel active application completely using pure PSO heavily for colour palette optimisation:** Pure PSO has strictly been actively used completely for pure codebook design previously. However, a strictly clear and totally reproducible pure implementation is completely provided entirely by this exact work. The strict absolute tradeoff completely balancing pure MSE reduction (sitting exactly $\approx 30\%$ strictly better entirely than pure KMeans heavily for $K = 16$) heavily against pure computational absolute cost (strictly running two complete orders of absolute magnitude slower) is actively quantified completely. A strict absolute benchmark heavily targeting purely future swarmintelligence based absolute compression strict systems is actively served entirely by these exact pure results.

4. **Integration strictly combining pure fuzzy clustering completely with active neural backpropagation:** That strict fuzzy absolute memberships pulled heavily from pure GathGeva can be completely used entirely to actively weight strict error pure updates is heavily demonstrated completely by the pure FBPNN absolute hybrid. Highly stable pure convergence actively paired directly with strictly competitive pure compression absolute ratios (hitting exactly $5.97 : 1$) is completely actively led strictly to by this. Entirely new strict avenues completely targeting pure alternative fuzzyneural absolute hybrids entirely within pure image coding are heavily opened completely by this specific action.

5. **Opensource, strictly modular pure codebase:** Python is completely used entirely to strictly implement all pure active algorithms. Extensive pure visualisation heavily paired entirely with strict absolute logging is actively included completely. The pure active code is strictly heavily structured entirely to completely allow the active easy absolute substitution completely of pure components (e.g., actively replacing pure KMeans directly with strict PSO heavily for absolute palette pure generation). Purely further strict research completely alongside active educational pure absolute use is heavily facilitated entirely by this.

4.4 Achievement of Objectives

Recalling the strict objectives entirely set out completely directly in Chapter 1:

1. **Implement completely fundamental pure lossless coding absolute algorithms (Huffman, arithmetic, LZW):** Fully completely achieved. All three were entirely implemented, completely actively verified strictly for losslessness, and their pure absolute performance was heavily evaluated entirely on strictly standard pure test images.

2. **Implement and entirely evaluate pure spatial absolute domain predictors (JPEG Lossless, LOCOI, CALIC):** Completely achieved. Pure LOCOI paired heavily with a strictly simplified pure CALIC were actively implemented; absolute JPEG Lossless (mode 4) entirely served heavily as a strict baseline. The active pure absolute advantage completely tied to strict context adaptation is heavily completely clearly shown entirely by the pure results.

3. **Implement perfectly pure integertointeger absolute wavelet transforms (S Transform, S+P, lifting):** Entirely achieved. All three strict absolute transforms were completely heavily implemented, entirely actively verified strictly for perfectly pure reconstruction, and their pure absolute entropy reduction was entirely heavily measured completely.

4. **Implement strictly lossy absolute colour pure quantisation methods (KMeans, GathGeva) paired directly with pure metaheuristic strict optimisation (PSO):** Completely achieved. Standard KMeans and pure GathGeva were actively implemented entirely; strict PSO was completely successfully entirely applied actively to strictly palette pure optimisation, heavily completely

demonstrating its absolute pure ability entirely to strictly actively find lowerMSE absolute solutions completely than standard KMeans.

5. **Develop an entirely hybrid pure Fuzzy Backpropagation Neural Network strictly for pure compression:** Heavily achieved. The pure absolute network was completely trained strictly upon pure Lenna, actively entirely achieving a strictly pure absolute compression ratio exactly hitting 5.97 : 1 completely paired directly with strictly highly stable absolute convergence entirely.

6. **Provide a strictly comprehensive comparative pure absolute analysis alongside complete guidelines heavily for algorithm absolute selection:** Entirely achieved. The strict pure absolute tradeoffs are heavily actively synthesised entirely by the pure active discussion completely within this chapter, and strictly clear absolute pure recommendations are heavily entirely provided completely for strictly absolute different pure use active cases (e.g., purely archival, strictly realtime, entirely highquality pure lossy).

4.5 Future Work

Several strictly clear absolute directions heavily pointing completely toward entirely extending this pure absolute work are actively identified completely below. **As noted in the scope statement at the beginning of this section, the actual implementation of these directions will be carried out in the next project (next semester). This report completes only the discovery and comparative analysis phase.**

- **Learned context models:** Handcrafted absolute pure context functions (e.g., pure gradient quantisation strictly inside LOCOI) could completely be actively replaced entirely by strictly small pure neural networks. Optimal pure predictors could be completely actively predicted strictly from pure absolute local neighbourhoods heavily by these. Residual pure entropy could absolutely be strictly reduced completely further by doing this, albeit completely at a strictly heavily higher absolute pure computational cost.

- **Adaptive transform coefficient selection:** Strictly completely fixed pure prediction absolute coefficients were heavily utilized completely by the pure S+P transform. Absolute pure energy compaction actively paired completely with strict entropy reduction could completely heavily be actively improved strictly by a pure learningbased or completely adaptive absolute coefficient selection mechanism (e.g., strictly utilizing pure PSO or entirely absolute gradient descent).

- **Perceptually guided colour quantisation:** Standard pure MSE could completely be actively replaced strictly entirely by absolute perceptual pure metrics (SSIM, MSSSIM, or purely CIEDE2000) completely within the strict pure fitness absolute functions heavily tied entirely to KMeans and pure PSO. Palettes that strictly heavily preserve absolute pure visual quality completely better specifically for the entirely absolute exact same pure K would be completely actively produced heavily by this exact action.

- **Scaling to larger datasets and realtime applications:** The strictly implemented pure PSO and absolute FBPNN mechanisms must be completely actively optimised heavily (e.g., strictly utilizing pure GPU absolute acceleration completely via JAX or pure PyTorch). This is strictly required entirely to actively handle purely highresolution absolute images and completely strictly achieve absolute pure realtime performance. Strict early absolute stopping completely partnered heavily with pure adaptive swarm sizing could completely be actively explored specifically for strict PSO.

- **Endtoend learned compression:** The strict pure FBPNN absolute concept could be completely heavily extended entirely into a strictly full pure autoencoder completely featuring absolute pure quantisation and strict entropy coding. This would be completely heavily trained jointly entirely upon a strictly massive pure image absolute dataset. The absolute pure massive strengths strictly tied entirely to pure fuzzy clustering, absolute neural prediction, and strictly entropy pure coding could completely be actively combined heavily into one strictly single pure absolute optimised system entirely by this.

- **Crossimage palette reuse:** A completely new strict pure palette does not actively strictly need to be completely computed entirely for every single pure image. Instead, a completely universal strict colour pure codebook can be actively completely learned strictly utilizing pure PSO or absolute GathGeva entirely upon a strictly diverse pure training set. This can then be actively completely reused strictly for the absolute pure compression completely tied to strictly similar pure images (e.g., heavily completely within pure medical or strictly satellite absolute imagery).

4.6 Final Remarks

A completely thorough strict pure empirical absolute comparison entirely spanning pure classic and completely modern pure image absolute compression strict algorithms has been heavily completely provided entirely by this absolute thesis. The completely critical strict absolute role entirely played actively by pure adaptivity – completely strictly whether entirely through pure contextbased active prediction, absolute pure transform energy strict compaction, or completely optimisationdriven strict palette pure design – is heavily actively highlighted entirely. That absolutely no strictly single pure algorithm completely entirely dominates heavily all pure absolute scenarios is strictly actively confirmed completely by the exact pure results. Rather, the absolutely optimal strict pure choice completely depends strictly heavily upon the explicitly specific pure absolute requirements tied completely to the pure application.

The strict pure lossyplusresidual absolute hybrid entirely stands completely out heavily as a purely absolute powerful strict paradigm completely for strict absolute lossless pure compression. Pure absolute ratios strictly completely unattainable entirely heavily by purely strict pure lossless methods are actively completely achieved entirely. Pure KMeans actively completely remains heavily the strict absolute workhorse specifically for pure speed completely heavily regarding entirely lossy pure colour strict quantisation. Superior pure absolute quality is completely heavily offered entirely by pure PSO strictly specifically when absolute pure time is completely actively permitted entirely. That strictly pure absolute hybrid pure approaches heavily completely act entirely as absolutely fertile strict ground completely for pure future absolute research is heavily actively suggested completely by the highly successful absolute pure integration strictly combining pure fuzzy absolute clustering heavily directly with strict neural pure backpropagation.

All strictly written pure code heavily alongside all absolute pure experimental strict data have been completely actively made entirely heavily available strictly. This strictly actively ensures complete pure absolute reproducibility and completely heavily serves strictly entirely as a pure absolute benchmark completely for strictly subsequent pure absolute work. That researchers heavily alongside active practitioners will completely be actively assisted heavily entirely in strictly completely selecting pure appropriate absolute compression pure techniques heavily entirely by this strictly absolute pure study is completely heavily hoped entirely. It is strictly also actively hoped completely that further absolute pure innovations strictly intersecting absolute pure machine learning, strict optimisation, and complete pure image coding will be heavily actively inspired completely entirely.

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